

MELETE — AI INTERVIEW PRACTICE PLATFORM

Comprehensive Technical, Architecture & Product Transformation Dossier

PROJECT: Melete Production SaaS	VERSION: 2.0 Enterprise SaaS	LIVE DEPLOYMENT: https://melete-seven.vercel.app
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1. Executive Summary & Evolution Overview

This technical dossier details the end-to-end transformation of Melete from a client-side frontend prototype into a high-scale, commercially viable SaaS platform. Designed specifically for international students preparing for university admissions and visa credibility interviews (Pre-CAS, UKVI, US F-1, Canada SDS, Germany APS, and Australia GS), Melete has been modernized to deliver automated real-time speech evaluation grounded in genuine regulatory criteria.

Key Core Accomplishment

Melete is now fully live on Vercel with real-time speech recording, OpenAI GPT-4o credibility grading, university-specific question banks, and an administrative control panel.

2. Benchmark Breakdown: Melete vs. ExamTestAI

Following our thorough inspection of ExamTestAI (a leading mock Pre-CAS platform for Nepali students applying to the UK), we benchmarked the systems to identify market-winning features and engineered Melete to exceed them:

Dimension	ExamTestAI (Market Competitor)	Melete (Our Production SaaS)
Scope	Narrow: Exclusively Nepali students applying to UK.	Broad International: UK, US F-1, Canada, Germany, Australia.
Speech Engine	Server-side audio capture & speech-to-text.	Real MediaRecorder + OpenAI Whisper-1 Audio Ingest.
AI Evaluation	AI Pre-CAS credibility analysis.	Real OpenAI GPT-4o-mini with regulatory red-flag triggers.
Feedback Model	"What we heard" + "Better way to say it" + Nepali.	Exact Transcript + Conversational English + Audio Playback.
Practice Modes	Full mock (17 Qs) or 1 question drill.	Both: Dedicated Single-Question Drill + Full 15-Q Mock.
Question Banks	UK university questions.	University Banks (Coventry, BPP, NYU, TUM) + Admin CRUD.
Pricing Model	Trial (10 Qs), then NPR 499 - 799 credit packs.	100% Free for Students with Google Auth session logging.
Admin Console	Proprietary internal console.	Commercial Dashboard: KPIs, Sessions, Transcripts & AI Config.

SYSTEM ARCHITECTURE & AI PIPELINE

Multi-Layer Technical Stack & Audio Ingest

3. End-to-End System Architecture

Melete was converted from pure static files into a robust, decoupled Node.js and Express platform designed for zero-fuss Vercel Serverless deployment:

```
Client (Browser)      % % %° MediaRecorder Audio Buffer + Canvas 60fps Spectrum
% (HTTPS POST multipart/form-data)
Vercel Serverless    % % %° Express API Router (api/index.js + server.js)
% % %° /api/transcribe % % %° OpenAI Whisper-1 Audio Transcription
% % %° /api/evaluate  % % %° OpenAI GPT-4o-mini Credibility Rubrics
```

4. Core AI & Audio Upgrades Implemented

- Real Audio Ingest & Dynamic Spectrum: Eliminated fake typewriter simulations. The client uses navigator.mediaDevices.getUserMedia and an AudioContext AnalyserNode to drive a 48-bar canvas visualizer reflecting candidate speech volume in real time.
- Audibility & Silence Guardian: Implemented an audio sanity check. If ambient silence or inaudible mumbles are detected (< 4 words), the system warns: "We couldn't hear your response clearly. Please check your mic and try again."
- OpenAI Whisper Audio Transcription: Uploaded student audio blobs (.webm/.wav) are processed by the Whisper API on the server, ensuring robust accent resilience across South Asian, African, and East Asian candidates.
- GPT-4o-mini Credibility Assessment: Student transcripts are evaluated against institutional rubrics. The AI outputs exact verbatim speech, a natural conversational English rewrite ("A better way to say it"), regulatory red flags, and single most important fixes.
- Intelligent Heuristic Fallback Engine: If external API keys or networks are temporarily unavailable, an internal linguistic-analysis engine runs automatically, guaranteeing 100% platform availability.

5. Dedicated Practice Modes

1. Drill 1 Question Mode (practice.html): Fast, focused single-question practice. Candidates pick country and university, record for 30–60 seconds, and get an instant AI evaluation with audio playback within 3 seconds.
2. Full 15-Question Mock Interview (session.html): Complete consular simulation with webcam streaming, timer, consecutive question progression, speech synthesis prompt delivery, and an end-of-session diagnostic report.

ADMINISTRATION & PRODUCTION DEPLOYMENT

Operational Governance & Live Vercel URLs

6. Commercial Admin Control Center (admin.html)

The Admin Dashboard gives administrators real-time oversight over student activity, system questions, and AI configuration:

Module	Capabilities & Features
Platform Overview	Live KPI metric cards: Registered Students, Total Mock Sessions, Average Credibility Score (%), and Stored Audio Files.
Session Vault	Inspect verbatim candidate transcripts, listen to student audio playback replays, and audit AI credibility grades.
Question Bank Manager	Full CRUD manager to add, edit, or delete questions, target universities, sample answers, and regulatory red flags.
Student Directory	Database records of registered candidates with Google authentication details, target countries, and session counts.
AI & System Settings	Manage OpenAI API keys (with secure masking), switch models (gpt-4o-mini, gpt-4o), and toggle Whisper transcription.

7. Live Production Verification & Telemetry

Melete was deployed live to Vercel production. The serverless functions, database layer, static assets, and OpenAI API endpoints are fully active and verified:

Production URLs

- Official Website: <https://melete-seven.vercel.app>
- Admin Dashboard: <https://melete-seven.vercel.app/admin.html>
- Drill 1 Question Mode: <https://melete-seven.vercel.app/practice.html>
- Full Mock Interview: <https://melete-seven.vercel.app/session.html>

8. Verified Live AI Response Telemetry

The following response was produced live by the production Vercel endpoint (<https://melete-seven.vercel.app/api/evaluate>) running the configured OpenAI API key:

```
HTTP/2 200 OK – content-type: application/json
{
  "score": 85, "status": "PASS",
  "what_heard": "I chose Coventry University because the MSc curriculum provides specialized modules...",
  "better_way": "I picked Coventry University because they offer great modules like Cloud Security...",
  "strengths": ["Specific mention of specialized modules", "Reference to Beatrice Shilling building"],
  "quick_tip": "Add more personal reasons for your choice to show genuine interest!"
}
```


